

The Relationship between Connectivity and Diagnosability on Regular Networks

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Abstract

The node connectivity is usually used to measure the fault tolerance of a multiprocessor system. On the other hand, the diagnosability is an important ability to provide a system level self-diagnosis for maintaining the reliability of a multiprocessor system. It is interesting to investigate the relationship between the connectivity and the diagnosability. Two diagnosis models: PMC model and comparison model are studied in this work. Herein, we consider that, under certain condition, a t -connected network is a t -diagnosable and is also a strongly t -diagnosable network. From our results, we can conclude that the diagnosability is enhanced as the connectivity improves for a multiprocessor system.

1 Introduction

A large number of topologies have been proposed and studied for multiprocessor systems. Such a topology is usually modeled as an undirected graph where the set of nodes represents the processors and the set of edges represents the communication links between the processors. In designing a multiprocessor system, one fundamental consideration is fault tolerance. One way to measure the fault tolerance of a system is based on the (node) connectivity of the underlying graph. In a system with connectivity t , the fault-free node is guaranteed to communicate with any other fault-free ones even if $(t-1)$ faulty nodes occur; in other words, the system keeps on connecting in the occurrence of faults. On the other hand, the complexity of the system can adversely affect the reliability when the number of processors in a multiprocessor system increases. To maintain the reliability of a system, the system should be able to discriminate the faulty nodes from the faulty-free ones, and then, the faulty nodes have to be replaced by fault-free nodes. The process of determining faulty nodes is called the *diagnosis* of the system. The maximum number of faulty nodes identified by the system is called the

diagnosability of the system. Motivated by these observations, the relationship between the connectivity and the diagnosability is studied in this paper.

For the purpose of self-diagnosis of a given system, several different models have been proposed in the literature [14], [15], and [16]. Preparata et al. [16] first introduced a model, called PMC-model, for system level diagnosis in multiprocessor systems. Under PMC model, a processor can test the faulty or fault-free status of adjacent processors. The PMC model was also used in [6], [9], [11], [13], and [19]. The comparison model, called MM model, is proposed by Malek and Maeng [14], [15]. The MM model deals with the fault diagnosis by sending the same input (or task) from a node w to each pair of distinct neighbors u and v , and then comparing their responses. The node w is called the *comparator* of the two nodes u and v . To obtain as much knowledge as possible about the faulty status of the system, it was assumed that a comparison is performed by each processor for each pair of distinct neighbors. This special case of the MM-model is referred to as the MM*-model. Different comparators may examine the same pair of nodes. The result of the comparison is that either the two responses are consistent or two responses disagree. The goal is to use the comparison results to identify the faulty/ fault-free status of the nodes in the system. Based on the comparison model, Sengupta and Dahbura characterized the diagnosable system and presented a polynomial algorithm to determine the set of all faulty nodes under the MM*-model [17]. The MM*-model was employed in [3], [8], [12], [20], and [21]. The PMC model and comparison model are studied in this work.

Given a system, it is impossible to identify that a processor v is fault-free or faulty when all the neighbors of the processor v are faulty simultaneously. Since most practical systems are sparsely connected, only a small number of faulty processors can be determined with the classical diagnosis model. Therefore, Lai et al. [13]

proposed two new measures of diagnosability where one is *strongly t -diagnosable* and the other is *conditional diagnosability* for multiprocessor systems. Given a t -diagnosable system, if all the neighbors of a processor v are not faulty simultaneously, the number of $t+1$ faulty node can be identified in the system under PMC model. This may be a practical situation in real system. In [13], the authors proved that under PMC model Matching Composition Network is a strongly t -diagnosable system and the conditional diagnosability of an n -dimensional hypercube is $4(n-2)+1$ for $n>4$. Furthermore, under the comparison model a t -regular and t -connected system may not be t -diagnosable and vice versa. Thus, it is interesting to investigate that, under certain condition, a given t -regular and t -connected system is a t -diagnosable system and is also a strongly t -diagnosable system. Herein, it will be specified the relationship between the connectivity and the diagnosability of multiprocessor systems under PMC model and the comparison model. Based on these results, we find many well-known multiprocessor systems which are t -diagnosable under PMC model and the comparison model. Moreover, they are strongly t -diagnosable systems with certain condition under PMC model and the comparison model.

In the next section, gives necessary backgrounds and notations used in this paper. The relationship between the connectivity and the diagnosability under different models are derived in Section 3. Section 4 presents the relationship between the connectivity and the diagnosability of some well-known multiprocessor systems under different models. Finally, we make concluding remarks in Section 5.

2 Preliminaries and notations

Let $G = (V, E)$ be a graph. We use V and E to represent the node set and edge set of G , respectively. The topology of a multiprocessor system is usually denoted by a graph $G = (V, E)$, where nodes represent processors and edges represent links between processors. Let V' be a subset of V , we use $G - V'$ to denote the subgraph of G induced by $V - V'$. The neighbor set of a node v is defined as $N(v) = \{u \in V \mid (u, v) \in E\}$. The neighbor set of a node v in V' is denoted by $N_{V'}(v)$. The connectivity of G is defined as $\kappa(G) = \min\{|V'| \mid V' \subseteq V \text{ and } G - V' \text{ is not connected}\}$. A graph G is t -connected if $\kappa(G) \geq t$. Given a t -connected graph, it follows from Menger's theorem that there exist t

internally node-disjoint (abbreviated as *disjoint*) paths between any two distinct nodes. On the other hand, for connectivity κ it has been also shown (Theorem 2.6 in [1]) that there exist κ disjoint paths from one node to another κ distinct nodes in a network.

Under the PMC model, adjacent nodes are capable of performing tests on each other. The ordered pair (u, v) represents the test performed by u on v . The node u is called the *tester* and v is called the *tested* node. The outcome of a test (u, v) is 1 (respectively, 0) if u evaluates v as faulty (respectively, fault-free). It can be modeled as a directed graph $T = (V, L)$ where an arc directed from node u to node v represents that u can test v . Let $F \subseteq V$ be the faulty set. The test result of node v by node u is denoted as $r(u, v)$ such that $r(u, v) = 0$ if u and $v \in V - F$ and $r(u, v) = 1$ if $u \in V - F$ and $v \in F$. The complete result of all tests for a test assignment T , defined as a function $S : L \rightarrow \{0, 1\}$, is called the *syndrome* of the diagnosis. On the other hand, the comparison scheme of the system can be modeled with a multigraph $M = (V, C)$, where V represents the node set and C the labeled-edge set. Let $(u, v)_w$ denote an edge labeled by w . In M , each $(u, v)_w \in C$ represents the node u and v being compared by w . The same pair of nodes may be compared by different comparators, so M is a multigraph. For $(u, v)_w \in C$, we use $r((u, v)_w)$ to denote the results of comparing nodes u and v by w such that $r((u, v)_w) = 0$ if the outputs of u and v agree and $r((u, v)_w) = 1$ if the outputs of u and v disagree. Clearly, if $r((u, v)_w) = 0$ and w is fault-free, then both u and v are fault-free. If $r((u, v)_w) = 1$, then at least one of u, v and w must be faulty. If w is faulty, then the result of comparison is unreliable in the sense that the exact status of u and v are unknown. The complete result of all comparisons, defined as a function $S : C \rightarrow \{0, 1\}$, is called the *syndrome* of the diagnosis.

A subset $F \subseteq V$ is said to be *consistent* with a syndrome S if S can arise from the circumstance that all the nodes in F are faulty and all the nodes in $V - F$ are fault-free. A system is said to be *diagnosable* if there is a unique $F \subseteq V$ that is consistent with S for every syndrome S . A system is called a *t -diagnosable* system if the system is diagnosable as long as the number of faulty nodes in it is no greater than t . Let $\sigma(F)$ denote the set of syndromes which could be produced if F is the set of faulty nodes. Two distinct sets $S_1, S_2 \subset V$ are said to be *indistinguishable* if

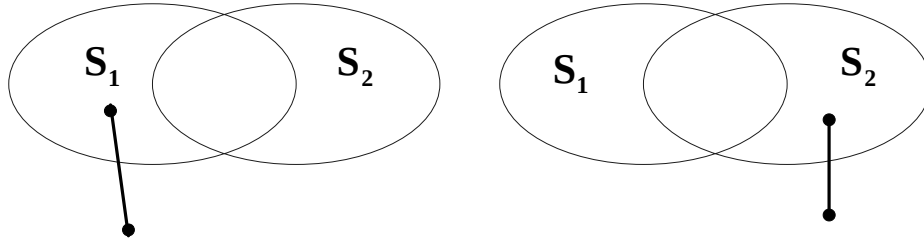


Figure 1. A distinguishable pair under PMC model.

and only if $\sigma(S_1) \cap \sigma(S_2) \neq \emptyset$; otherwise, S_1 and S_2 are said to be *distinguishable*. Let $S_1 \Delta S_2 = (S_1 - S_2) \cup (S_2 - S_1)$ be the symmetric difference.

The following two lemmas are useful to the distinguishability of two sets under PMC model and comparison model, respectively. Lemma 2 will be used later to prove a strongly t -diagnosable system under comparison model, respectively.

Lemma 1 [6]. A system $G = (V, E)$ is t -diagnosable under PMC model if and only if, for each pair $S_1, S_2 \subset V$ with $|S_1|, |S_2| \leq t$ and $S_1 \neq S_2$, there is at least one test from $V - (S_1 \cup S_2)$ to $S_1 \Delta S_2$ (as shown in Fig. 1).

Lemma 2 [16]. For any S_1, S_2 where $S_1, S_2 \subset V$ and $S_1 \neq S_2$, (S_1, S_2) is a distinguishable pair if and only if at least one of the following conditions is satisfied: (as illustrated in Fig. 2)

- (1) $\exists u, w \in V - S_1 - S_2$, and $\exists v \in S_1 \Delta S_2$ such that $(u, v)_w \in C$,
- (2) $\exists u, v \in S_1 - S_2$, and $\exists w \in V - S_1 - S_2$, such that $(u, v)_w \in C$,
- (3) $\exists u, v \in S_2 - S_1$, and $\exists w \in V - S_1 - S_2$, such that $(u, v)_w \in C$.

Definition 1. A system G is strongly t -diagnosable if the following two conditions hold:

1. G is t -diagnosable and
2. For any two distinct subsets $S_1, S_2 \subset V$ with $|S_i| \leq t+1, i=1, 2$, either
 - a. (S_1, S_2) is a distinguishable pair or
 - b. (S_1, S_2) is an indistinguishable pair and there exists a node $v \in V$ such that $N(v) \subseteq S_1$ and $N(v) \subseteq S_2$.

According to the above definition, a $(t+1)$ -diagnosable system is strongly t -diagnosable and it is naturally “stronger” than a t -diagnosable system. However, among all the strongly t -diagnosable systems, we are interested in one which is t -diagnosable rather than $(t+1)$ -diagnosable. The necessary and sufficient condition proved by Lai et al. [13] is employed to prove a strongly t -diagnosable system under PMC

model, and the result is listed in the following lemma.

Lemma 3 [21]. A system is strongly t -diagnosable under PMC model if and only if, for each node set $S \subset V$ with $|S| = p$, $0 \leq p \leq t$, the following two conditions are satisfied:

1. for $0 \leq p \leq t-1$, every component C of $G - S$ satisfies $|V_c| \geq 2((t+1)-p)+1$ and
2. for $p = t$ either (a) every component C of $G - S$ satisfies $|V_c| \geq 3$ or else (b) $G - S$ contains at least one trivial component.

3 Relationship between Connectivity and Diagnosability on Regular Networks

In this section, a t -regular and t -connected network with $N \geq 2t+3$ nodes is shown to be a t -diagnosable system under PMC model and the comparison model. Furthermore, the problems are also considered that, under certain condition, a t -regular and t -connected network are also strongly t -diagnosable under PMC model and the comparison model, respectively. Two theorems were respectively proven in [3] and [9] listed in the following.

Theorem 1 [9]. Let G be a t -regular and t -connected network with N nodes. G is t -diagnosable under PMC model if $N \geq 2t+1$.

Theorem 2 [3]. Let G be a t -regular and t -connected network with N nodes and $t > 2$. G is t -diagnosable under the comparison model if $N \geq 2t+3$.

It follows from Theorems 1 and 2 that a t -regular and t -connected network with at least $2t+3$ nodes is t -diagnosable under both PMC model and comparison model.

By Lemma 3, the strongly t -diagnosable system is almost $(t+1)$ -diagnosable under PMC model. The only case that stops it from being $(t+1)$ -diagnosable occurs in the following situation: All the neighbors of a node v are faulty simultaneously.

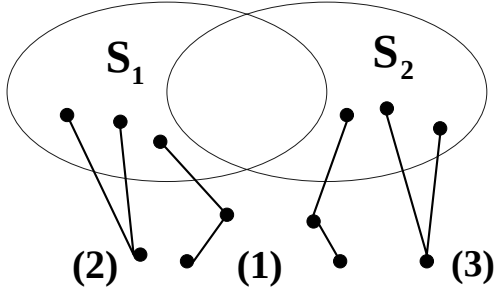


Figure 2. Three conditions of distinguishable under comparison model

Herein, it is shown that given a t -regular and t -connected network with at least $2t+3$ nodes is also strongly t -diagnosable under PMC model if the system is triangle free and there is at least one neighbor not in common for each pair of distinct nodes u and v in the system.

Theorem 3. For $t > 2$, let G be a t -regular and t -connected network with $N \geq 2t+3$ nodes. If G is triangle-free and each pair of distinct nodes u and v of G has $N(u) \neq N(v)$, G is strongly t -diagnosable under PMC model.

Proof. Let $S \subset V$ and $|S| = p$. Since $G - S$ is connected with only one connected component for $0 \leq p \leq t-1$, every connected component C satisfies $|V_C| = N - p \geq 2t+3 - p$. Thus, the condition 1 of Lemma 3 holds. For $p = t$, since $G - S$ may be disconnected, the following cases are considered.

Case 1. $G - S$ has at least two trivial components.

Let v_1 and v_2 be two trivial components of $G - S$. It follows that $N(v_1) \subseteq S$ and $N(v_2) \subseteq S$. Since $N(v_1) \neq N(v_2)$ and $|N(v_1) \cap N(v_2)| \leq t-1$, we have $|S| \geq 2t - (t-1) = t+1$ which is a contradiction for $|S| = t$. Thus, it is impossible that $G - S$ has at least two trivial components.

Case 2. There exists one nontrivial components R with $|R| = 2$ in $G - S$.

Let v_1 and v_2 be two nodes of component R . It follows that there are $t-1$ neighbors of v_1 and v_2 in S , respectively. Since $|S| = t$ and $2(t-1) > t$ for $t > 2$, it must exist at least one common neighbor v_c of v_1 and v_2 in S . Hence, three nodes v_1, v_2 , and v_c is a cycle of length 3 (i.e., triangle). It is a contradiction that G is triangle-free.

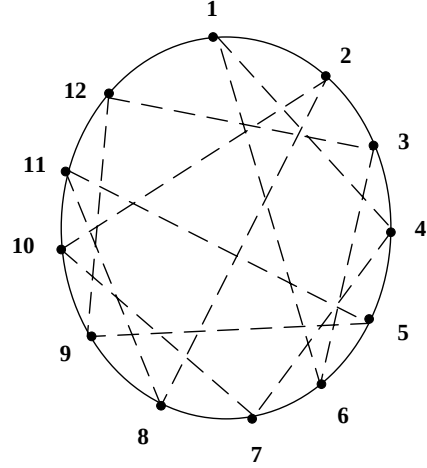


Figure 3. An example of 4-connected and nonstrongly 4-diagnosable with 12 nodes.

Thus, there does not exist any nontrivial component R with $|R| = 2$ in $G - S$.

Then let v be a node of V and $N(v) = S$. It follows that $G - S$ is a disconnected graph and there exists a trivial component v . Thus, the condition 2 of Lemma 3 is satisfied. Furthermore, the other nontrivial components have the number of nodes at least 3 in $G - S$ such that G is almost $(t+1)$ -diagnosable except all the neighbors of a node v are faulty simultaneously. Therefore, the theorem is complete. $\square$

Now, given a t -regular and t -connected network $G = (V, E)$ with $N \geq 2t+3$ nodes, we consider that G is strongly t -diagnosable under comparison model. Let $S_1, S_2 \subset V$ with $|S_i| \leq t+1$, $i = 1, 2$ and let $S_1 \cap S_2 = S$ with $|S| = p$, $0 \leq p \leq t$. Since G is t -diagnosable, all possible choices of S_1 and S_2 with $|S_i| \leq t$, $i = 1, 2$ need not be considered. Furthermore, there are three examples shown in the following which are t -regular and t -connected networks without triangle but they are not strongly t -diagnosable under comparison model. The graph shown in Fig. 3 is a 4-regular and 4-connected network without triangle but it is not strongly 4-diagnosable under comparison model. There are $N = 2t + 4 = 12$ nodes. Let $S_1 = \{1, 7, 11, 5, 9\}$ and $S_2 = \{1, 7, 11, 2, 3\}$. The pair (S_1, S_2) is an indistinguishable pair under comparison model. Similarly, the Fig. 4 is a 5-regular and 5-connected network with $N = 2t + 4 = 14$ nodes and triangle-free. Let $S_1 = \{5, 7, 9, 11, 13, 14\}$ and $S_2 = \{2, 3, 5, 7, 9, 11\}$. It is easy to see that the pair (S_1, S_2) is an indistinguishable pair under comparison model. Also, the Fig. 5 is a 6-regular and 6-connected network with $N = 2t + 4 = 16$ nodes and triangle-free.

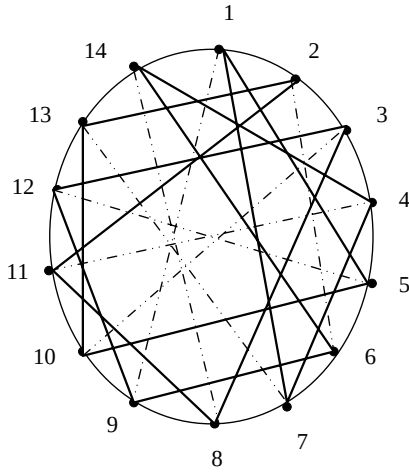


Fig. 4. An example of 5-connected and nonstrongly 5-diagnosable with 14 nodes.

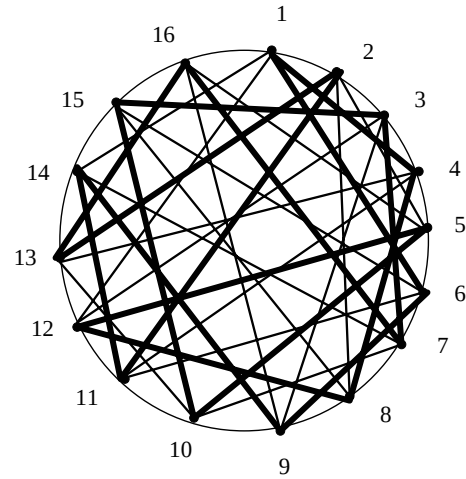


Fig. 5. An example of 6-connected and nonstrongly 6-diagnosable with 16 nodes

Let $S_1 = \{1, 5, 8, 9, 11, 13, 15\}$ and $S_2 = \{1, 3, 5, 7, 11, 13, 15\}$. It is easy to see that the pair (S_1, S_2) is an indistinguishable pair under comparison model.

Hence, the t -regular and t -connected network G with $N \geq 2t+3$ nodes for $t \geq 7$ is considered in the following lemmas.

Lemma 4. Given a t -regular and t -connected network G with $N \geq 2t+3$ nodes for $t \geq 7$. Let $S_1, S_2 \subset V$ with $|S_i| = t+1$, $i=1, 2$ and let $S_1 \cap S_2 = S$ with $|S| = p$, $0 \leq p \leq t-1$. (S_1, S_2) is a distinguishable pair under comparison model if G is triangle-free.

Since $G - S$ may not be connected for $|S| = p = t$, the following lemma is distinguished from Lemma 4 to be individually considered.

Lemma 5. For $t \geq 7$, given a t -regular and t -connected network G with $N \geq 2t+3$ nodes, let $S_1, S_2 \subset V$ with $|S_i| = t+1$, $i=1, 2$ and let $S_1 \cap S_2 = S$ with $|S| = p$, $p = t$. (S_1, S_2) is a distinguishable pair under comparison model if the following two conditions hold:

1. G is triangle-free.
2. Each pair of distinct nodes u and v of G has $|N(u) \cap N(v)| = k$ with $k \leq t-2$.

By Lemmas 4 and 5, we have the following theorem that a t -regular and t -connected network G with $N \geq 2t+3$ nodes and $t \geq 7$ is strongly t -diagnosable under comparison model.

Theorem 4. For $t \geq 7$, given a t -regular and t -connected network G with $N \geq 2t+3$ nodes. G is strongly t -diagnosable under comparison model if the following conditions are

satisfied:

1. G is triangle-free.
2. Each pair of distinct nodes u and v of G has $|N(u) \cap N(v)| = k$ with $k \leq t-2$.

Proof. Let $S_1, S_2 \subset V$ with $|S_i| \leq t+1$, $i=1, 2$ and let $S_1 \cap S_2 = S$ with $|S| = p$, $0 \leq p \leq t$. It follows from Theorem 2 that G is t -diagnosable such that the condition 1 of definition 1 holds. Since G is t -diagnosable but it is not $(t+1)$ -diagnosable, (S_1, S_2) is a distinguishable pair for $|S_i| \leq t$, $i=1, 2$ and $0 \leq p \leq t-1$. For $|S_i| = t+1$, $i=1, 2$ and $0 \leq p \leq t$, it follows from Lemmas 4 and 5 that (S_1, S_2) is a distinguishable pair. Hence, the condition 2(a) of definition 1 is satisfied.

Now, let $v \in V$ and $S = N(v)$. Assume that $S_1 = S \cup v$ and $S_2 = S$. Thus, we have $|S_1| = t+1$, $|S_2| = t$ and $|S| = p = t$. There exists a node $v \in V$, $N(v) \subseteq S_1$, and $N(v) \subseteq S_2$ such that (S_1, S_2) is an indistinguishable pair. The condition 2(b) of definition 1 is satisfied. Therefore, the theorem holds. $\square$

4 Application to Regular Networks

Four theorems in Section 3 are applied to five popular regular networks while these four theorems are also possible to apply them to many other potentially useful ones not shown in here.

Example 1 [17]. The n -dimensional Hypercube Q_n is recursively constructed as follows: Q_1 is a complete graph with two nodes labeled by 0 and 1, respectively. Q_n consists of two identical subcubes Q_{n-1}^0 and Q_{n-1}^1 . The pair of nodes u

and v is an edge in Q_n if and only if $H(u, v) = 1$.

The following is the definition of the n -dimensional Crossed cube. Two binary strings $x = x_1x_0$ and $y = y_1y_0$ are pair related if and only if $(x, y) \in \{(00, 00), (10, 10), (01, 11), (11, 01)\}$, denoted by $x \sim y$.

Example 2 [7]. [10]. The n -dimensional Crossed cube CQ_n is recursively constructed as follows: CQ_1 is a complete graph with two nodes labeled by 0 and 1, respectively. CQ_n consists of two identical subcubes CQ_{n-1}^0 and CQ_{n-1}^1 . The pair of nodes $u \in CQ_{n-1}^0$ and $v \in CQ_{n-1}^1$ is an edge in CQ_n if and only if

1. $u_{n-2} = v_{n-2}$ if n is even, and
2. $u_{2i+1}u_{2i} \sim v_{2i+1}v_{2i}$ for $0 \leq i < \lfloor (n-1)/2 \rfloor$.

The n -dimensional Twisted cube is defined by $P_i(u)$ for n odd, where $P_i(u) = (u_i + u_{i-1} + \dots + u_0) \bmod 2$.

Example 3 [10]. The n -dimensional Twisted cube TQ_n is recursively constructed as follows: TQ_1 is a complete graph with two nodes labeled by 0 and 1, respectively. TQ_n consists of four identical subcubes TQ_{n-2}^{00} , TQ_{n-2}^{01} , TQ_{n-2}^{10} and TQ_{n-2}^{11} . For $0 \leq i < (n-1)/2$, edges connect these four subcubes described as follows.

1. $u_{2i}u_{2i-1} = \bar{v}_{2i}v_{2i-1}$ and $u_{2i}u_{2i-1} = \bar{v}_{2i}\bar{v}_{2i-1}$ if $P_{2i-2}(u) = 0$,
2. $u_{2i}u_{2i-1} = \bar{v}_{2i}v_{2i-1}$ and $u_{2i}u_{2i-1} = v_{2i}\bar{v}_{2i-1}$ if $P_{2i-2}(u) = 1$,
3. $u_{2j}u_{2j-1} = v_{2j}v_{2j-1}$ for all $j \neq i$, where u_0 is the single bit for $i = 0$.

Let TQ_{n-1}^0 (TQ_{n-1}^1 , respectively) be the subgraph of TQ_n induced by $TQ_{n-2}^{00} \cup TQ_{n-2}^{10}$ ($TQ_{n-2}^{01} \cup TQ_{n-2}^{11}$, respectively). Then TQ_n is constructed from TQ_{n-1}^0 and TQ_{n-1}^1 by connecting these two subcubes with a specific perfecting matching. Next, we define the Möbius cubes in the following. There are two types of Möbius cubes MQ_n , namely, 0- MQ_n and 1- MQ_n which can be rewritten as follows.

Example 4 [5]. The n -dimensional Möbius cube MQ_n is recursively constructed as follows: Both 0- MQ_1 and 1- MQ_1 are complete graph with two nodes labeled by 0 and 1, respectively. Both 0- MQ_n and 1- MQ_n consist of one 0-type MQ_{n-1}^0 subcubes and one 1-type MQ_{n-1}^1 subcubes. The pair of nodes $u = 0u_{n-1}u_{n-2} \dots u_{i+1}u_iu_{i-1} \dots u_0$ and v is an edge in 0- MQ_n with $v = 0u_{n-1}u_{n-2} \dots 0\bar{u}_iu_{i-1} \dots u_0$ or $v = 0u_{n-1}u_{n-2} \dots 1\bar{u}_iu_{i-1} \dots \bar{u}_0$ for $0 \leq i \leq n-1$. The pair of nodes $u = 1u_{n-1}u_{n-2} \dots u_{i+1}u_iu_{i-1} \dots u_0$ and v is an edge in 1- MQ_n with $v = 1u_{n-1}u_{n-2} \dots 0\bar{u}_iu_{i-1} \dots u_0$ or

$v = 1u_{n-1}u_{n-2} \dots 1\bar{u}_i\bar{u}_{i-1} \dots \bar{u}_0$ for $0 \leq i \leq n-1$.

Example 5 [1]. A star graph of n dimensions can be expressed by a graph $S_n = (V, E)$ with V being the set of all permutations of $\{1, 2, \dots, n\}$ and E consisting of all (u, v) such that $u = u_1u_2 \dots u_k \dots u_n$ and $v = u_ku_2 \dots u_{k-1}u_1u_{k+1} \dots u_n$ (i.e., swap u_1 and u_k) for $2 \leq k \leq n$.

Since these five regular networks G are triangle-free and $|N(u) \cap N(v)| \leq 2$ for each pair of distinct nodes u and v of G , by Theorems 1 and 2, the Q_n , CQ_n , TQ_n , and MQ_n are n -diagnosable system under PMC model and comparison model. Similarly, by Theorems 1 and 2, the S_n is $(n-1)$ -diagnosable system under PMC model and comparison model. Furthermore, by Theorem 3, the Q_n , CQ_n , TQ_n , and MQ_n are strongly n -diagnosable system and the S_n is strongly $(n-1)$ -diagnosable system under PMC model. Also, by Theorem 4, the Q_n , CQ_n , TQ_n , and MQ_n are strongly n -diagnosable system and the S_n is strongly $(n-1)$ -diagnosable system under comparison model.

5 Conclusion

In this paper, the relationship between the connectivity and the diagnosability of a multiprocessor system is studied. It is shown that the multiprocessor system is n -diagnosable under PMC model and the comparison model if an n -regular and n -connected multiprocessor system has at least $2n+3$ nodes. Furthermore, given an n -regular and n -connected multiprocessor system with at least $2n+3$ nodes is strongly n -diagnosable under PMC model if the system is triangle free and there is at least one neighbor not in common for each pair of distinct nodes u and v in the system. Similarly, an n -regular and n -connected multiprocessor system with at least $2n+3$ nodes is also strongly n -diagnosable under the comparison model if the system is triangle free and there are $k \leq n-2$ common neighbors for each pair of distinct nodes u and v in the system and $n \geq 7$. Thus, we can conclude the diagnosability is enhanced as the connectivity improves for a multiprocessor system.

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